

SpaceX Falcon 9 First Stage Landing Prediction

Data Science Capstone Project Report

IBM Applied Data Science Capstone · Coursera

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GitHub: <https://github.com/pankh-ui/IBM-Applied-Data-Science-Capstone>

Project Repository

All notebooks, the dashboard script and this report

<https://github.com/pankh-ui/IBM-Applied-Data-Science-Capstone>

- 1-jupyter-labs-spacex-data-collection-api.ipynb — SpaceX REST API data collection
- 2-jupyter-labs-webscraping.ipynb — BeautifulSoup scraping of the Wikipedia launch table
- 3-labs-jupyter-spacex-Data-wrangling.ipynb — landing outcome converted to a binary label
- 4-jupyter-labs-eda-dataviz.ipynb — exploratory charts and one-hot feature engineering
- 5-jupyter-labs-eda-sql-coursera_sqlite.ipynb — the ten SQL exploration queries
- 6-lab-jupyter-launch-site-location.ipynb — Folium site, outcome and proximity maps
- 7-SpaceX-Machine-Learning-Prediction.ipynb — four tuned classifiers and their evaluation
- spacex_dash_app.py — the Plotly Dash launch records dashboard
- Data_Science_Capstone_Project_Report.pdf — this presentation

Outline

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 - Data wrangling — cleaning, landing-outcome labeling, one-hot encoding
 - Exploratory data analysis with visualization and with SQL
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Executive Summary

Methods used and the key results, in brief

- Goal: predict whether the Falcon 9 first stage will land successfully, so that the cost of a launch can be estimated. A reused first stage takes a launch from roughly \$165M to about \$62M.
- Methodologies used:
 - Data collection from the SpaceX REST API (JSON → pandas) and web scraping of the Falcon 9 launch table on Wikipedia with BeautifulSoup
 - Data wrangling: null handling, filtering to Falcon 9, and converting eight landing outcomes into a binary Class label (1 = landed, 0 = did not land)
 - EDA with pandas / Matplotlib / Seaborn and with SQL (SQLite magic commands)
 - Interactive visual analytics with Folium maps and a Plotly Dash dashboard
 - Predictive analysis: Logistic Regression, SVM, Decision Tree and KNN, each tuned with GridSearchCV (10-fold CV) on standardized features
- Key results:
 - Launch success has risen steadily from 0% in 2010 to roughly 80% by 2019-2020
 - KSC LC-39A has the highest success rate of any site; ES-L1, GEO, HEO and SSO orbits show 100% success in this dataset
 - Low-weight payloads (< 4000 kg) land successfully more often than heavy ones
 - All four models reach the same test accuracy of 83.33%; the Decision Tree has the best validation score (86.25%), and the confusion matrix shows false positives are the only meaningful error mode

Introduction

Project background and the problem being solved

- **Background:**

- SpaceX advertises Falcon 9 launches at \$62 million, while other providers charge upward of \$165 million. The saving comes from re-using the first stage.
- If the first stage does not land, the booster is lost and that launch costs far more. Predicting the landing therefore predicts the cost of the launch.
- Space Y, a competitor, wants to bid against SpaceX and needs that cost model.

- **Problems we want to answer:**

- What factors determine whether the first stage lands successfully?
- How do payload mass, launch site, orbit type and booster version interact with the landing outcome?
- Has the landing success rate improved over time?
- Which classification model best predicts a successful landing, and how accurate is it?

Methodology

How the data was collected, cleaned, explored and modeled

Methodology — Overall Workflow

- 1. Data collection — SpaceX REST API (api.spacexdata.com/v4/launches/past) and web scraping of the Wikipedia Falcon 9 launch records table.
- 2. Data wrangling — filter to Falcon 9, impute missing PayloadMass with the column mean, and convert the eight landing outcomes into a binary Class label.
- 3. Exploratory data analysis — Seaborn / Matplotlib charts and SQL queries (SQLite via %sql magic) to find the relationships that matter.
- 4. Interactive visual analytics — Folium maps of every launch site and their proximities, plus a Plotly Dash dashboard with a site dropdown and a payload slider.
- 5. Predictive analysis — standardize features, split 80/20, tune Logistic Regression, SVM, Decision Tree and KNN with GridSearchCV (cv=10), then compare accuracy and confusion matrices on the held-out test set.

Data Collection Methodology — SpaceX API

Key phrases and the flow of the API request

- Flow: GET <https://api.spacexdata.com/v4/launches/past> → `response.json()` → `pd.json_normalize()` → helper calls → filter to Falcon 9 → impute nulls → `dataset_part_1.csv`
- `requests.get(spacex_url)` — retrieve the past-launch records as JSON.
- `pd.json_normalize(response.json())` — flatten the nested JSON into a dataframe.
- `getBoosterVersion`, `getLaunchSite`, `getPayloadData`, `getCoreData` — call the rocket, launchpad, payload and core endpoints by ID to replace the IDs with real values.
- Filter: `data_falcon9 = data[data['BoosterVersion'] != 'Falcon 1']` — 90 launches remain.
- Missing values: PayloadMass nulls replaced with the column mean; LandingPad keeps its 26 nulls because 'no landing pad' is meaningful.

Data Collection Methodology — Web Scraping

Scraping the Falcon 9 launch table from Wikipedia

- Flow: `requests.get(static_url)` → `BeautifulSoup(html, 'html.parser')` → `find_all('table')` → parse header cells → row-by-row extraction → dataframe → `spacex_web_scraped.csv`
- `static_url` — the 9 June 2021 snapshot of 'List of Falcon 9 and Falcon Heavy launches'.
- `soup.title` returns: List of Falcon 9 and Falcon Heavy launches - Wikipedia — confirming the correct page was fetched.
- `html_tables = soup.find_all('table')`; the third table holds the launch records.
- `extract_column_from_header(th)` builds the column names, giving Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date and Time.
- Each row is walked with `find_all('td')`, the helper functions `date_time()`, `booster_version()`, `landing_status()` and `get_mass()` clean each cell, and the dictionary `launch_dict` is converted with `pd.DataFrame(launch_dict)`.

Data Wrangling Methodology

Cleaning, exploring and labeling the landing outcome

- Missing values: PayloadMass nulls replaced with the mean of the column; LandingPad nulls retained (26) because they encode 'no landing pad used'.
- Exploratory counts: launches per site (CCAFS SLC 40 = 55, KSC LC 39A = 22, VAFB SLC 4E = 13), occurrences of each orbit, and mission outcome per orbit type.
- Landing outcome label — the Outcome column holds eight values combining the result with the landing place (True ASDS, True RTLS, True Ocean, False ASDS, False RTLS, False Ocean, None ASDS, None None).
- bad_outcomes = {False ASDS, False Ocean, False RTLS, None ASDS, None None}; every outcome in that set becomes Class = 0 and everything else becomes Class = 1.
- Result: a clean binary target. `df['Class'].mean()` = 0.667, i.e. a two-thirds overall success rate across the 90 launches.
- Categorical features (Orbit, LaunchSite, LandingPad, Serial) were one-hot encoded with `pd.get_dummies`, producing an 80-column feature matrix cast to float64.

EDA and Data Visualization — Methodology

Charts plotted and why each one was chosen

- Scatter — Flight Number vs Payload Mass: shows that success improves with flight number and that very heavy payloads are rare.
- Scatter — Flight Number vs Launch Site: reveals which site handled the early (mostly failing) flights and which handled the later, more successful ones.
- Scatter — Payload Mass vs Launch Site: shows that VAFB SLC 4E has no launches above 10 000 kg.
- Bar — success rate by orbit type: identifies the orbits with perfect and with poor landing records.
- Scatter — Flight Number vs Orbit and Payload vs Orbit: separates a genuine orbit effect from an effect that is really about experience or payload weight.
- Line — yearly average success rate: the trend over time, which is the single clearest signal in the data.
- Scatter plots were chosen for pairwise relationships with a class color, bar charts for comparing a rate across categories, and a line chart for the time trend.

EDA with SQL

Summary of the SQL queries performed

- Displayed the names of the unique launch sites in the space mission.
- Displayed 5 records where the launch site name begins with 'CCA'.
- Displayed the total payload mass carried by boosters launched by NASA (CRS).
- Displayed the average payload mass carried by booster version F9 v1.1.
- Listed the date of the first successful landing outcome on a ground pad.
- Listed the boosters that had success on a drone ship and carried a payload between 4000 kg and 6000 kg.
- Listed the total number of successful and failed mission outcomes.
- Listed the booster versions that carried the maximum payload mass, using a subquery.
- Listed the failed drone-ship landings in 2015 with booster version, launch site and the month name.
- Ranked the count of landing outcomes between 2010-06-04 and 2017-03-20 in descending order.

Interactive Visual Analytics

Folium map and Plotly Dash — objects and why

- Folium — folium.Circle and folium.Marker with a DivIcon label mark every launch site, so the geographic spread of the sites is immediately visible.
- Folium — MarkerCluster groups the 56 launch records at their sites; each marker is colored green for success and red for failure, so a site's record can be read at a glance without over-plotting.
- Folium — folium.PolyLine plus a haversine distance function measures the distance from CCAFS SLC-40 to the coastline, railway, highway and nearest city, testing whether sites keep to safe proximities.
- Dash — dcc.Dropdown (site-dropdown) selects All Sites or one specific site.
- Dash — a pie chart callback shows either the share of successful launches per site or the success-versus-failure split for the selected site.
- Dash — dcc.RangeSlider (payload-slider, 0-10 000 kg) plus a scatter callback shows payload against landing outcome, colored by booster version category.

Predictive Analysis (Classification) — Methodology

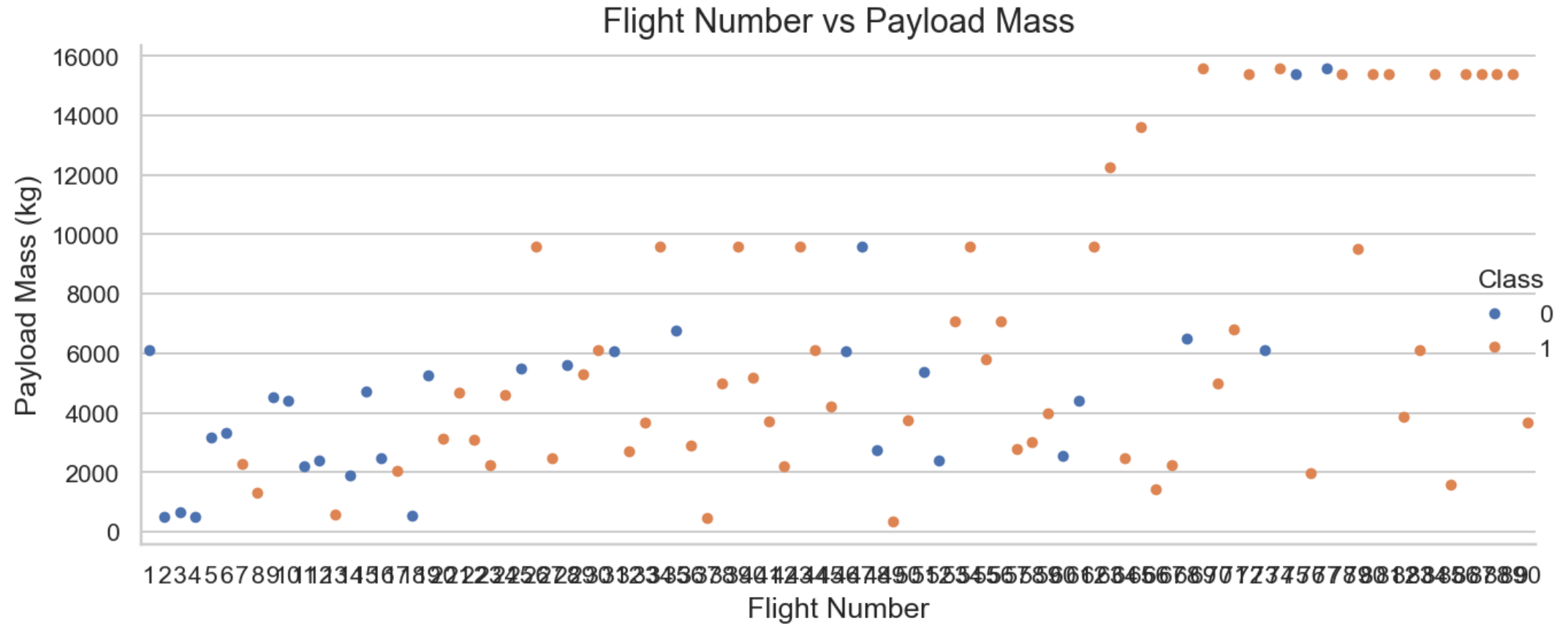
- Target: `Y = df['Class']` (1 = first stage landed, 0 = did not land).
- Features: the 80-column one-hot encoded matrix, standardized with `StandardScaler().fit_transform(X)`.
- Split: `train_test_split(X, Y, test_size=0.2, random_state=2)` → 72 training rows and 18 test rows.
- Four algorithms tuned with `GridSearchCV(cv=10)`:
 - Logistic Regression — `C`, `penalty`, `solver`
 - Support Vector Machine — `kernel` (linear, rbf, poly, sigmoid), `C`, `gamma`
 - Decision Tree — `criterion`, `splitter`, `max_depth`, `max_features`, `min_samples_leaf`, `min_samples_split`
 - K-Nearest Neighbours — `n_neighbors`, `algorithm`, `p`
- Evaluation: `best_score_` for the validation accuracy, `.score(X_test, Y_test)` for the test accuracy, and a confusion matrix for the error profile of the best model.

Results

EDA, SQL, interactive analytics and predictive modelling

Flight Number vs Payload Mass

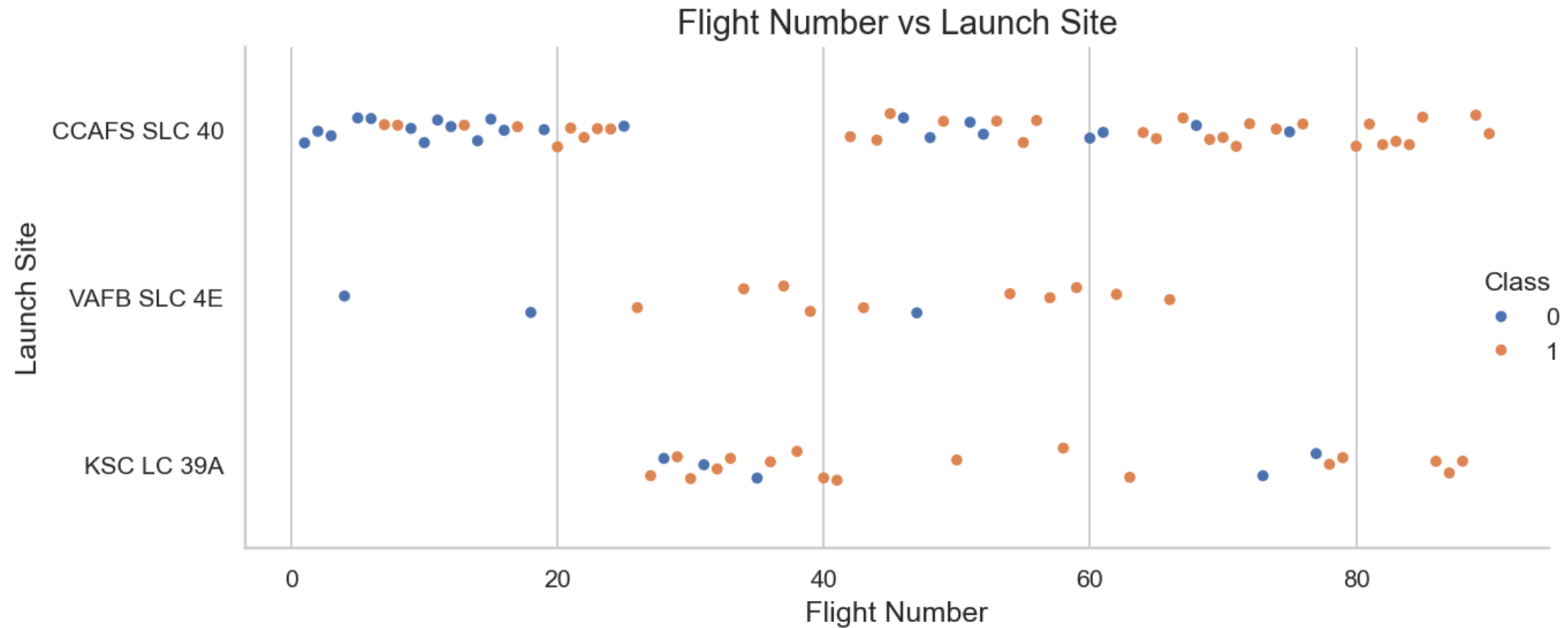
EDA with Visualization — Results



Successes (orange) dominate at higher flight numbers. Payloads above 10 000 kg are rare, and the heaviest ones still land successfully.

Flight Number vs Launch Site

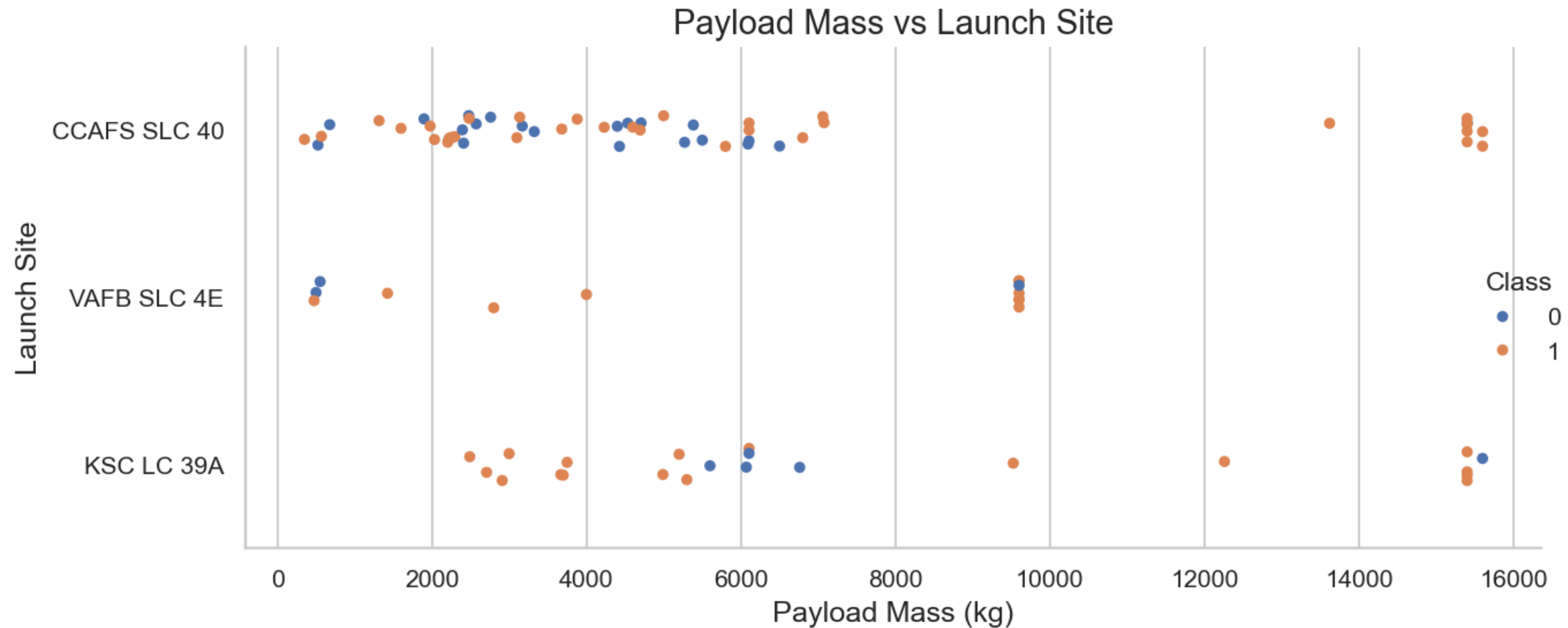
EDA with Visualization — Results



CCAFS SLC 40 carries most of the early failures; KSC LC 39A only appears after flight 25 and is almost entirely successful.

Payload Mass vs Launch Site

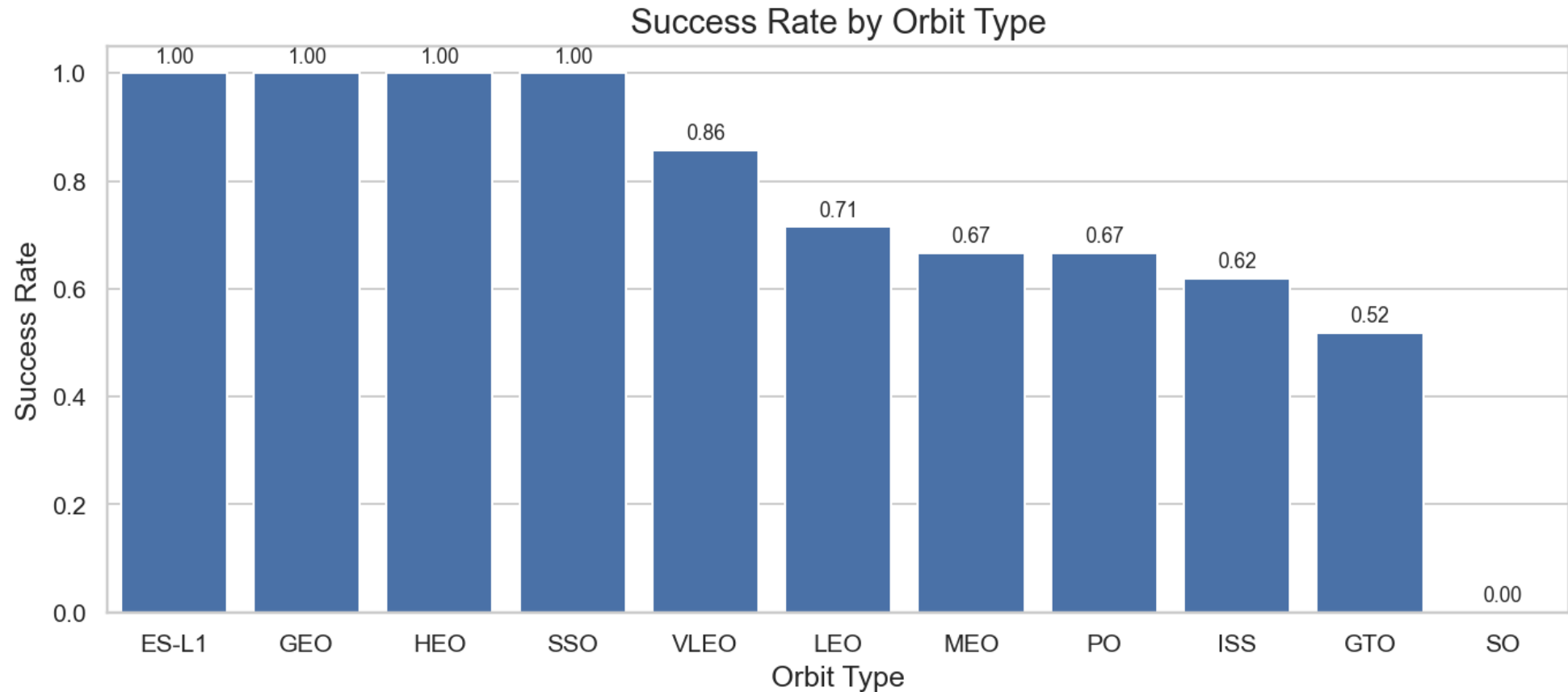
EDA with Visualization — Results



VAFB SLC 4E has no launch above 10 000 kg. Above about 7 000 kg the success rate improves at the Florida sites.

Success Rate by Orbit Type

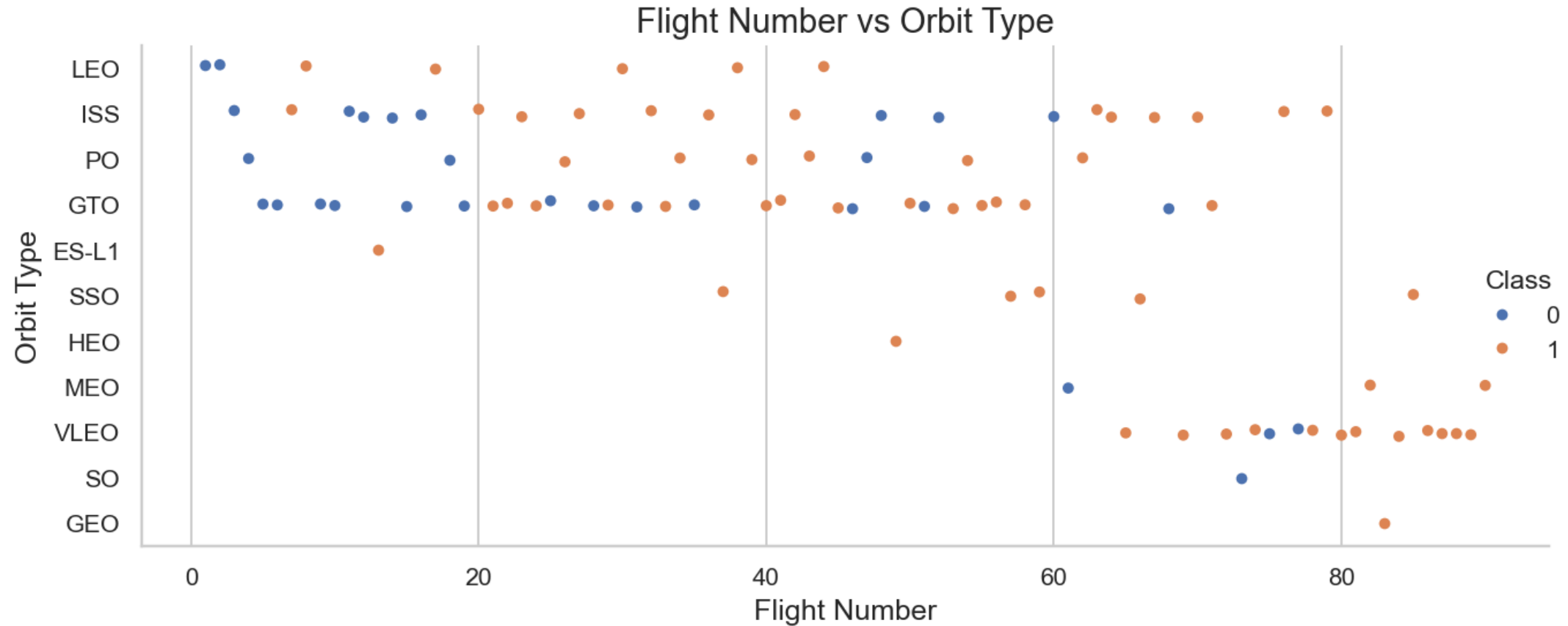
EDA with Visualization — Results



ES-L1, GEO, HEO and SSO show a 100% success rate in this dataset; SO shows 0%. GTO, the busiest orbit, sits around 0.5.

Flight Number vs Orbit Type

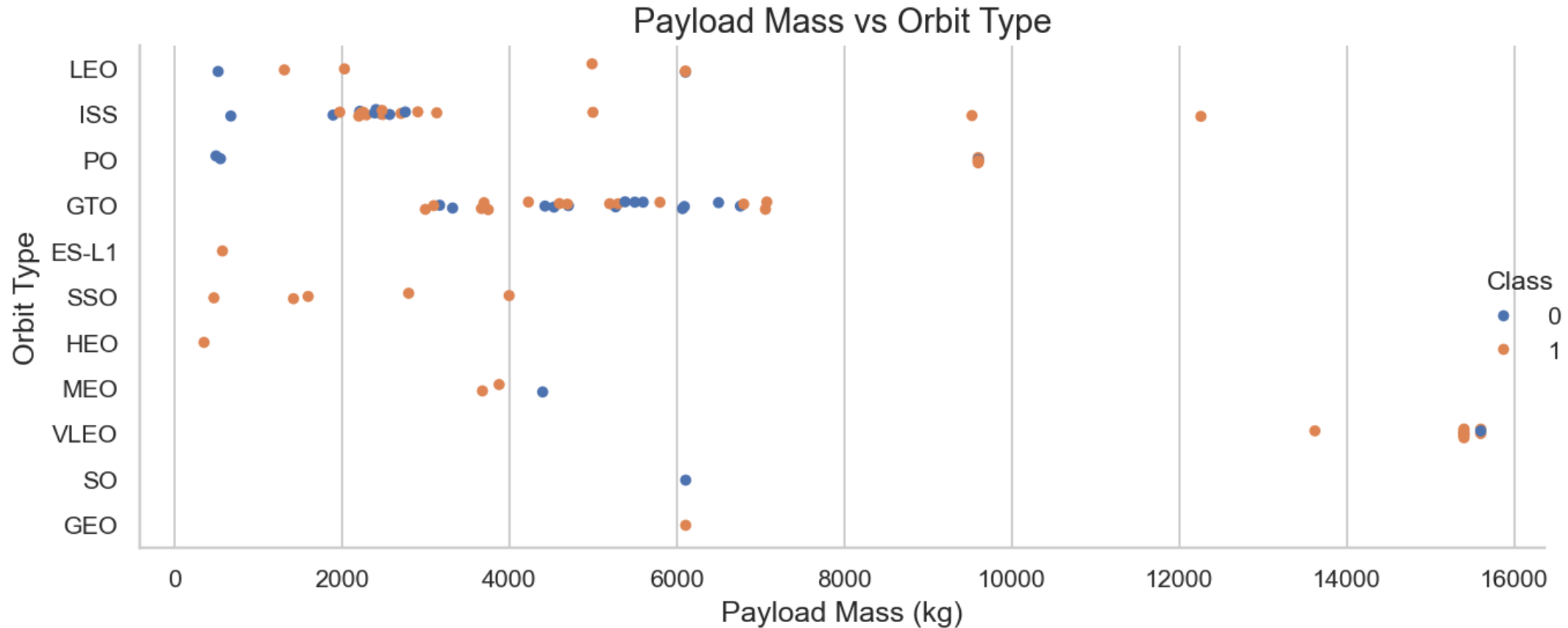
EDA with Visualization — Results



In LEO, success is clearly related to flight number — experience matters. In GTO there is no such relationship.

Payload Mass vs Orbit Type

EDA with Visualization — Results



Heavy payloads land more successfully in LEO, ISS and PO. For GTO the picture is mixed, so payload alone does not explain the outcome.

Launch Success Yearly Trend

EDA with Visualization — Results



Success was 0% until 2013 and then climbed steadily to roughly 80% by 2019-2020, with a dip in 2018. This is the strongest single trend in the data.

All Launch Sites

EDA with SQL — Results and Queries

Task 1 — Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Four launch sites appear in the data set.

Launch Site Names Beginning with 'CCA'

EDA with SQL — Results and Queries

Task 2 — 5 records where launch site begins with 'CCA'

Date	Launch_Site	Booster_Version	Payload	PAYLOAD_MASS__KG_
2010-06-04	CCAFS LC-40	F9 v1.0 B0003	Dragon Spacecraft Qualification	0
2010-12-08	CCAFS LC-40	F9 v1.0 B0004	Dragon demo flight C1, two CubeSats, barre	0
2012-05-22	CCAFS LC-40	F9 v1.0 B0005	Dragon demo flight C2	525
2012-10-08	CCAFS LC-40	F9 v1.0 B0006	SpaceX CRS-1	500
2013-03-01	CCAFS LC-40	F9 v1.0 B0007	SpaceX CRS-2	677

The first five records from Cape Canaveral, showing early Falcon 9 v1.0 flights.

Total Payload Mass — NASA (CRS)

EDA with SQL — Results and Queries

Task 3 — Total payload mass carried by boosters from NASA (CRS)

Total_Payload_Mass_KG
45596

Total payload carried by boosters flown for NASA's Commercial Resupply Services.

Average Payload Mass — F9 v1.1

EDA with SQL — Results and Queries

Task 4 — Average payload mass carried by booster version F9 v1.1

Avg_Payload_Mass_KG
2928.4

The average payload for the F9 v1.1 booster version.

First Successful Ground-Pad Landing

EDA with SQL — Results and Queries

Task 5 — Date of the first successful landing outcome on ground pad

First_Successful_Ground_Landing
2015-12-22

The first successful landing on a ground pad, found with MIN(Date).

Successful Drone-Ship Landings, 4000-6000 kg

EDA with SQL — Results and Queries

Task 6 — Boosters: success on drone ship, payload 4000-6000 kg

Booster_Version	PAYLOAD_MASS__KG_
F9 FT B1022	4696
F9 FT B1026	4600
F9 FT B1021.2	5300
F9 FT B1031.2	5200

Boosters that landed on a drone ship while carrying a mid-weight payload.

Total Successful and Failed Mission Outcomes

EDA with SQL — Results and Queries

Task 7 — Total number of successful and failed mission outcomes

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Mission outcomes counted with GROUP BY — almost every mission succeeded, whatever happened to the booster.

Boosters Carrying the Maximum Payload

EDA with SQL — Results and Queries

Task 8 — Booster versions that carried the maximum payload mass

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

Found with a subquery on MAX(PAYLOAD_MASS__KG_) — all are B5 boosters.

2015 Failed Drone-Ship Landings

EDA with SQL — Results and Queries

Task 9 — 2015 failure landing outcomes on drone ship (month, booster, site)

Month	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Month name extracted with substr() and a CASE statement.

Landing Outcomes Ranked, 2010-06-04 to 2017-03-20

EDA with SQL — Results and Queries

Task 10 — Landing outcomes ranked, 2010-06-04 to 2017-03-20

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Counted and ordered descending — 'No attempt' dominates the early period.

Launch Sites on a Folium Map

Folium Map — Results



All four launch sites marked with a circle and a label. Every site is close to the coast and near the equator, which helps the rocket gain speed from the Earth's rotation and keeps failed launches over water.

Launch Records Clustered by Site

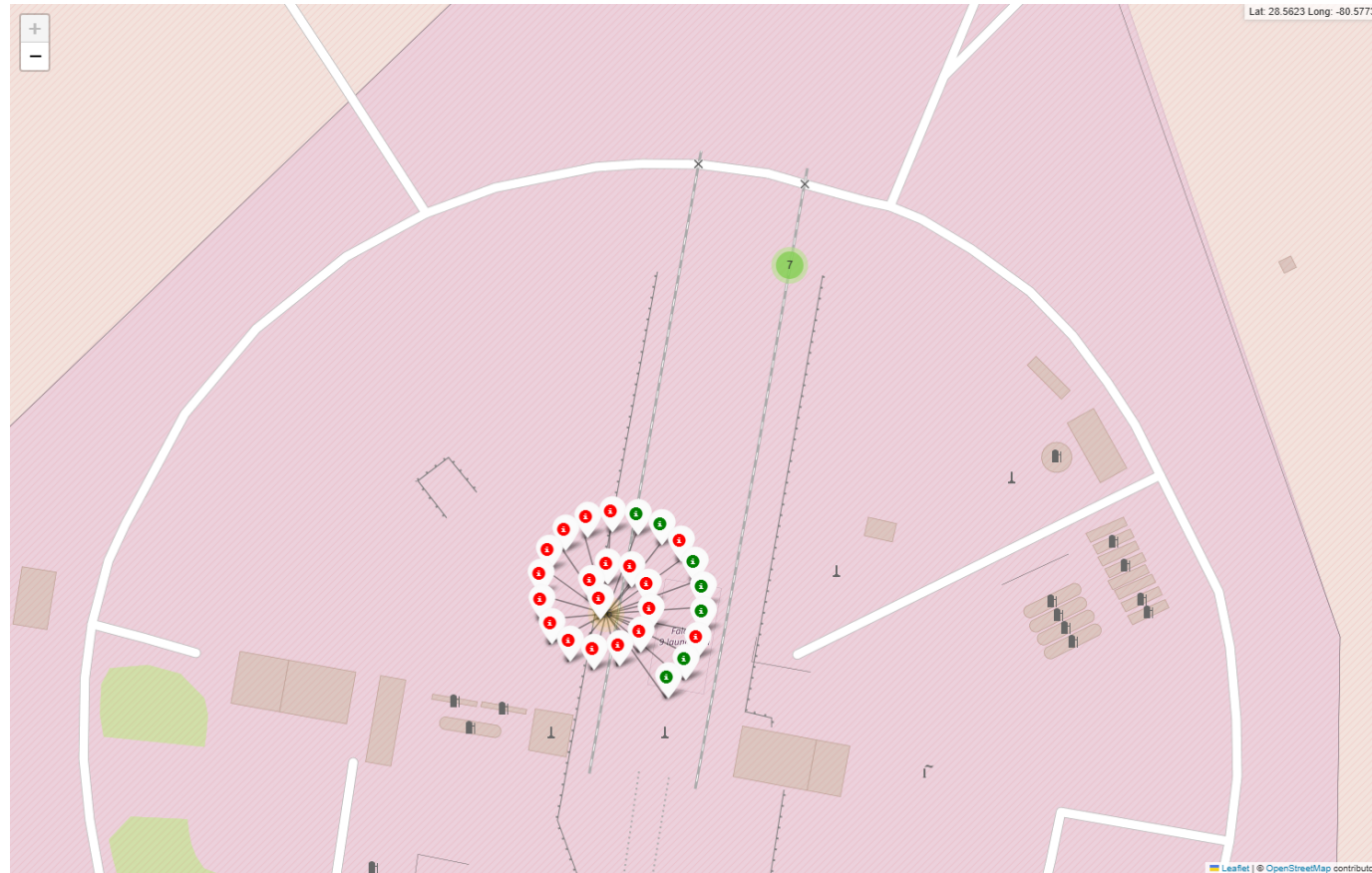
Folium Map — Results



MarkerCluster groups the launch records at each site, so the relative launch volume of the Florida sites versus Vandenberg is immediately clear.

Success and Failure Markers at CCAFS SLC-40

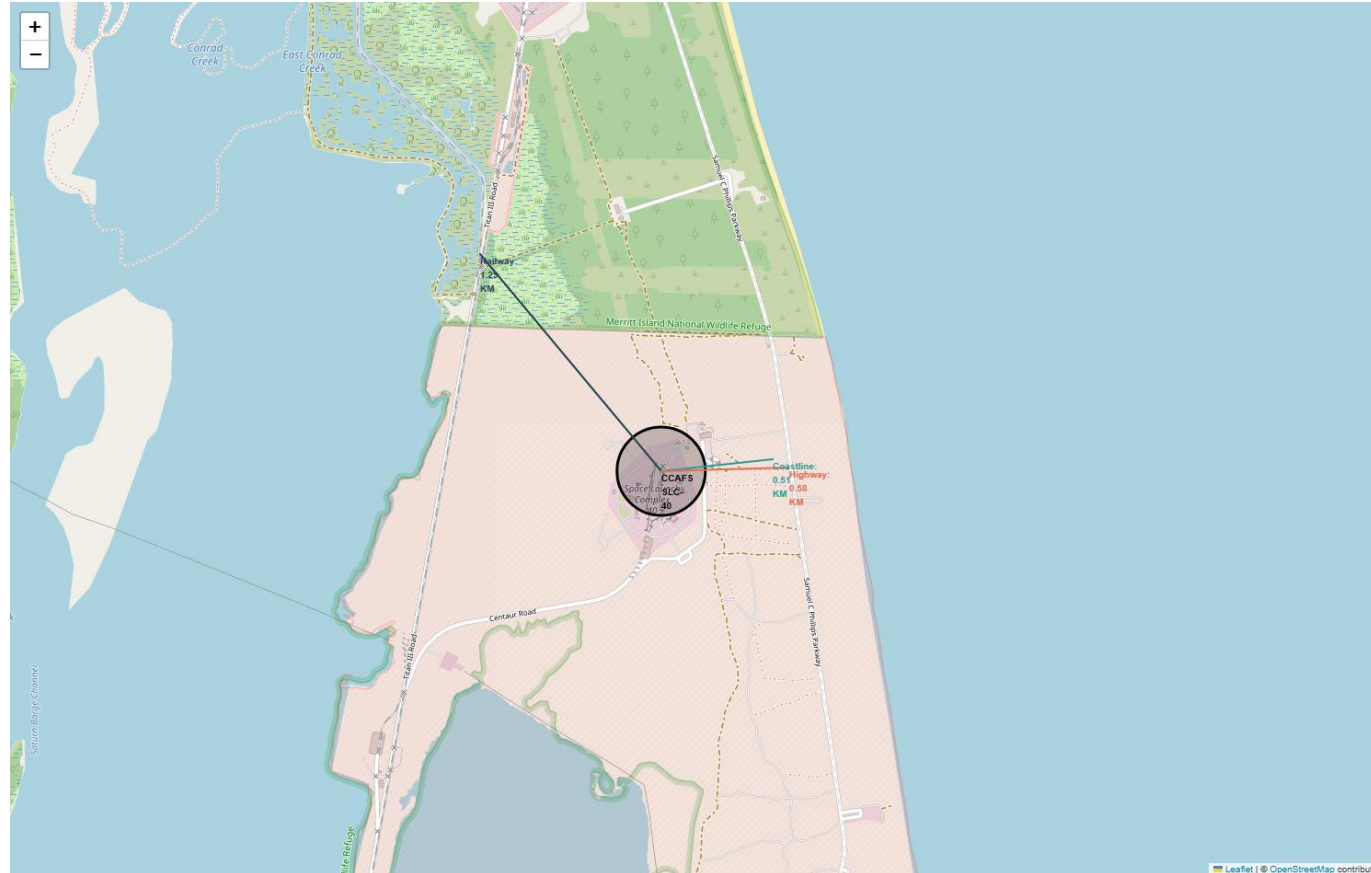
Folium Map — Results



Expanding the cluster shows each launch colored green (landed) or red (did not land). CCAFS SLC-40 has a clear majority of red markers, reflecting its many early flights.

Launch Site Proximities — CCAFS SLC-40

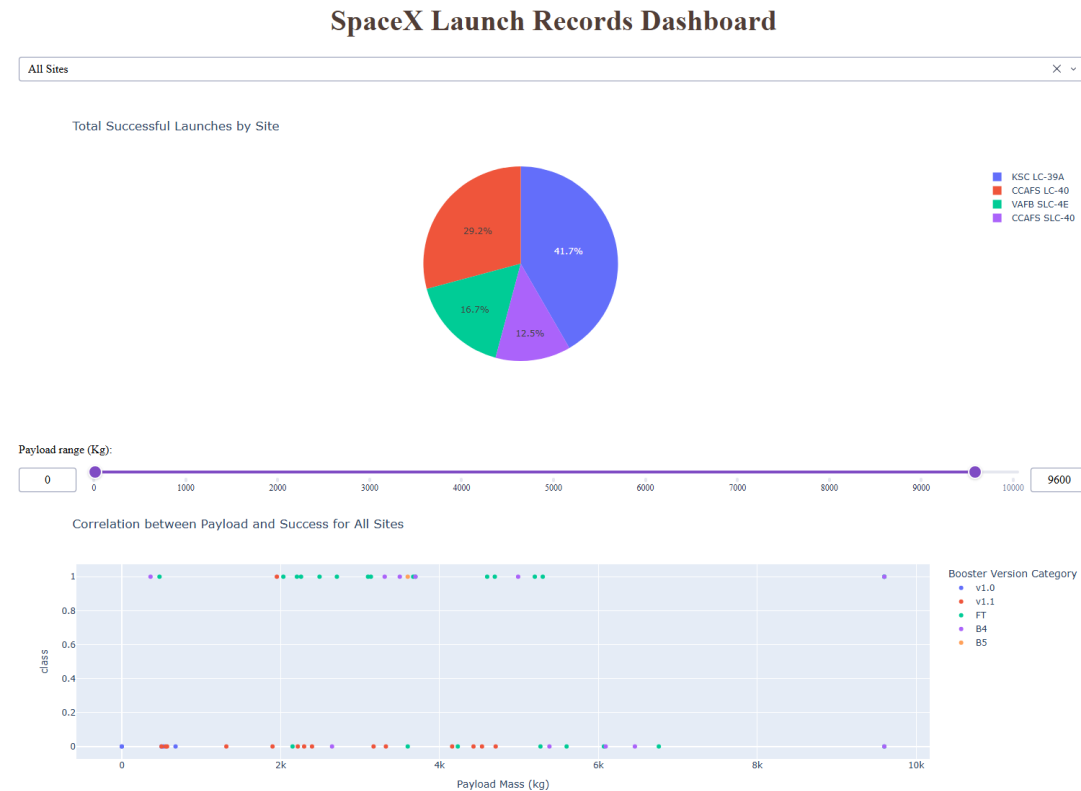
Folium Map — Results



Distances measured with a haversine function: coastline 0.51 km, highway 0.58 km, railway 1.29 km, and the nearest city (Titusville) 23.19 km. Sites sit close to transport links but keep well away from population centres.

Plotly Dash Dashboard — All Sites

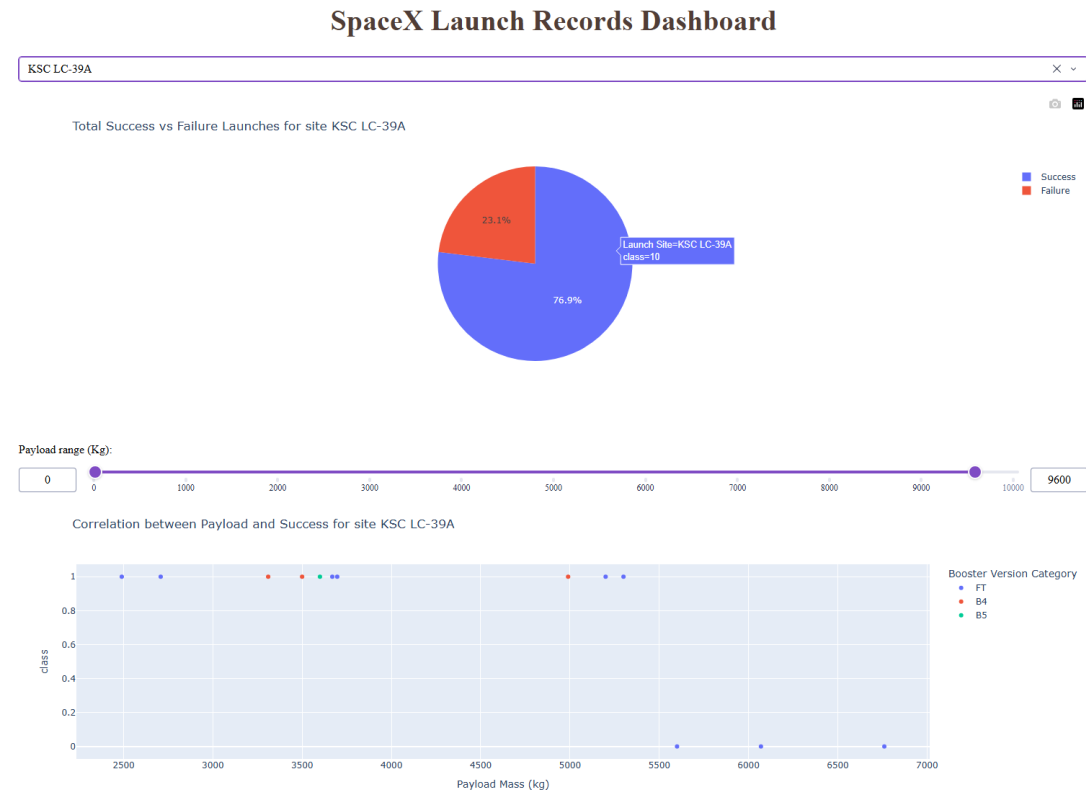
Plotly Dash Dashboard — Results



The pie chart shows the share of all successful launches by site: KSC LC-39A leads with 41.7%, followed by CCAFS LC-40 (29.2%), VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%). The scatter below shows payload against outcome for every site.

Plotly Dash Dashboard — KSC LC-39A Selected

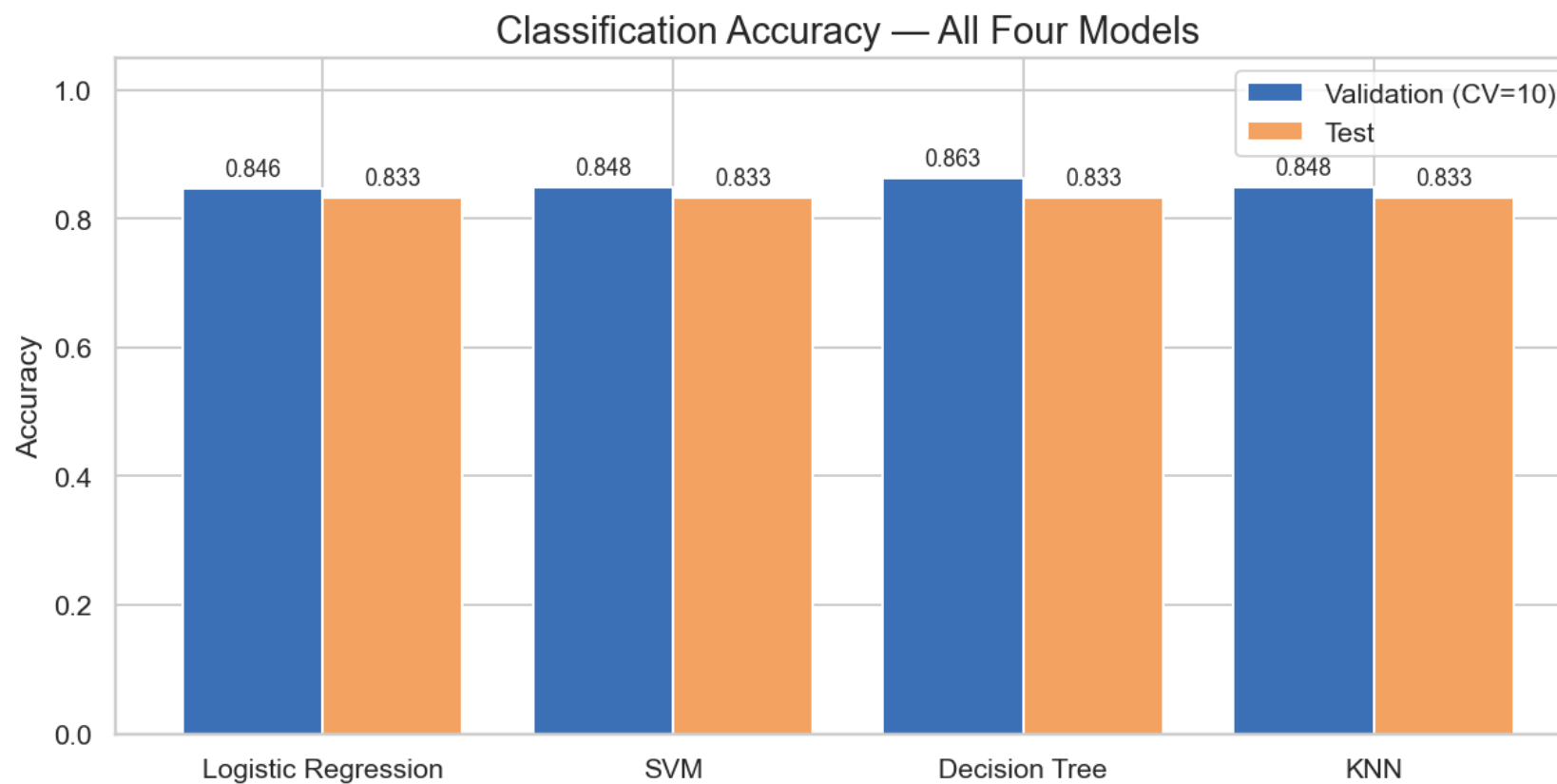
Plotly Dash Dashboard — Results



Selecting a single site switches the pie chart to that site's success-versus-failure split and filters the payload scatter. KSC LC-39A has the highest success ratio of any site, and its successes concentrate in the low payload range with FT and B4 boosters.

Classification Accuracy — All Models

Predictive Analysis (Classification) — Results



All four models reach exactly the same test accuracy of 83.33% (15 of 18 test launches). The Decision Tree wins on validation accuracy at 86.25%.

Model Comparison Table

Predictive Analysis (Classification) — Results

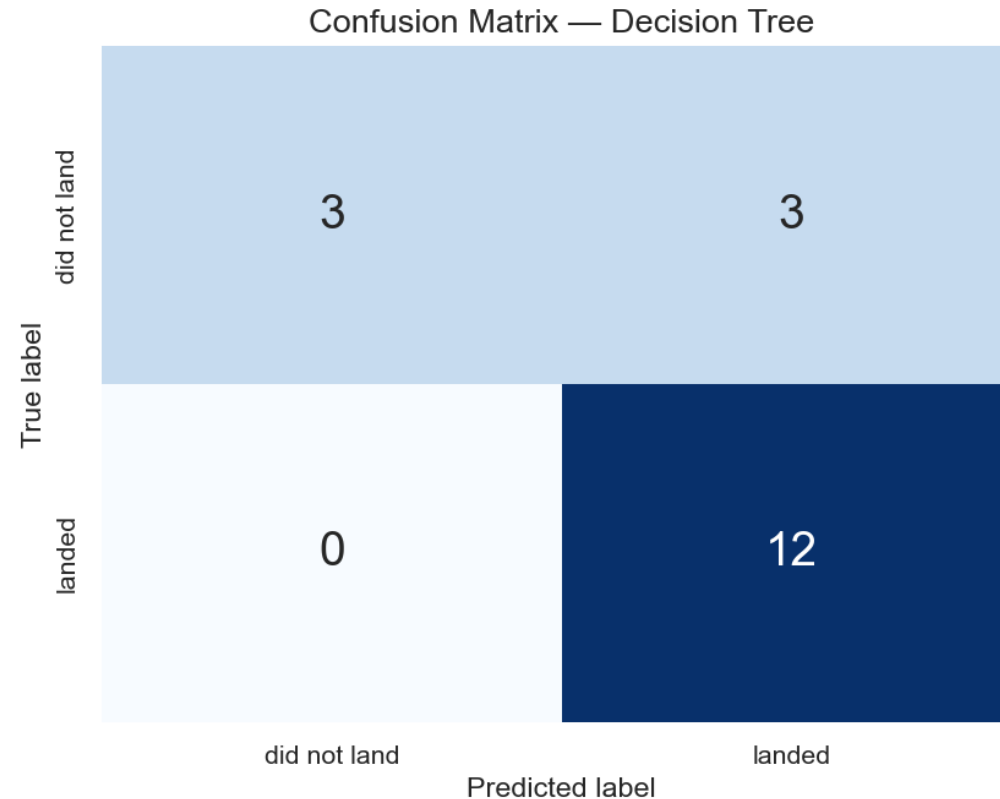
Model comparison — validation vs test accuracy

Model	Best validation accuracy (CV=10)	Test accuracy
Logistic Regression	0.8464	0.8333
SVM	0.8482	0.8333
Decision Tree	0.8625	0.8333
KNN	0.8482	0.8333

- Logistic Regression — best hyperparameters: {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
- SVM — best hyperparameters: {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}
- Decision Tree — best hyperparameters: {'criterion': 'gini', 'max_depth': 6, 'max_features': 'sqrt', 'min_samples_split': 2}
- KNN — best hyperparameters: {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}

Confusion Matrix — Decision Tree

Predictive Analysis (Classification) — Results



Of the 18 test launches the model correctly identifies all 12 that landed and 3 of the 6 that did not. There are no false negatives — the model never misses a successful landing — but three false positives, meaning it is optimistic about boosters that in fact fail. For a cost model that errs on the risky side, so the false-positive rate is the metric to improve.

Best Performing Model

- Best model: Decision Tree — validation accuracy 0.8625, test accuracy 0.8333.
- All four models score identically on the test set because the test set holds only 18 launches — a single reclassified launch moves accuracy by 5.6 percentage points. The validation score across 10 folds is the more reliable comparison.
- The Decision Tree is also the most interpretable of the four, which matters for a cost model that has to be explained to a bidding team.
- Caveat: with 90 rows and 80 features the models are working in a very wide, very shallow space. More launches, or aggressive feature selection, would firm up these results.

Conclusion

- Launch success improves with experience: the yearly success rate climbs from 0% in 2010-2013 to about 80% by 2019-2020.
- Launch site matters. KSC LC-39A has the best record; CCAFS SLC 40 carries the early failures simply because it flew the early missions.
- Orbit matters. ES-L1, GEO, HEO and SSO show 100% success in this dataset, while GTO — the busiest orbit — sits near 50%.
- Payload matters, but less than expected: lighter payloads (< 4000 kg) land successfully more often, yet the very heaviest payloads in LEO, ISS and PO still land.
- Prediction: with only 90 launches, all four tuned classifiers reach 83.33% test accuracy. The Decision Tree has the best cross-validated score and is the model to carry forward.
- Business meaning: a bidder can estimate launch cost from site, orbit, payload and booster version before the launch is flown — which is exactly what Space Y needs to price a competing bid.

Innovative Insights

What the data shows beyond the required analysis

- The strongest predictor is not physical but temporal. Flight number — a proxy for accumulated engineering experience — separates the classes better than payload mass does. SpaceX's landing problem was solved by iteration, not by flying easier missions.
- Orbit success rates are partly an artefact of volume. The orbits at 100% (ES-L1, GEO, HEO, SSO) each have only a handful of launches; GTO looks worse largely because it has the most attempts. Any orbit-based rule needs a sample-size caveat.
- The confusion matrix reveals an asymmetric risk. The model never misses a success but over-predicts three failures as successes, so a bid priced from this model would be too optimistic. Weighting the loss toward false positives would be the next iteration.
- The launch sites cluster tightly against the coast, close to a highway and railway but over 20 km from the nearest town — a geographic pattern that a competitor choosing a new site would have to replicate.

Appendix

- Notebooks: data collection (API), data collection (web scraping), data wrangling, EDA with visualization, EDA with SQL, Folium interactive map, and machine-learning prediction.
- Dashboard: `spacex_dash_app.py` — Plotly Dash application with a launch-site dropdown, a success pie chart, a 0-10 000 kg payload range slider, and a payload-versus-outcome scatter chart.
- Data sources: SpaceX REST API v4 (api.spacexdata.com/v4/launches/past) and the Wikipedia page 'List of Falcon 9 and Falcon Heavy launches' (9 June 2021 snapshot).
- Libraries: pandas, numpy, matplotlib, seaborn, beautifulsoup4, requests, sqlite3 / ipython-sql, folium, plotly, dash and scikit-learn.
- Dataset size after wrangling: 90 Falcon 9 launches, 80 features after one-hot encoding, 72 training rows and 18 test rows.